

1. Executive Summary

		Traffic Summary		Count	
Metric	Value	Total Flows Observed		16843	
Total Threats Mitigated	19	Total Packets Analyzed		3591531964	
High Priority	10	Malicious Packets		3579340155	
Low Priority	9	True Negatives Passed		12249	
Manual Releases	0				
Manual Blocks	0	Attack Vector		Count	
		ICMP Flood		0	
ML Metrics	Value	SYN Flood		0	
False Positives	24	UDP Flood		15	
FP Rate	0.20%	Uncertain		4	
IF Precision	99.47%				
IF Recall	97.67%	Action		Count	
IF F1-Score	98.56%	Quarantined		4	
IF Accuracy	99.21%	Time Ban		3	
		Blackhole		0	
		Released		3	

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	99.47%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	97.67%	Share of actual attacks successfully flagged
F1-Score	98.56%	Balanced measure combining Precision and Recall
Accuracy	99.21%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.20%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	2.33%	Actual attacks that went undetected
True Positive Rate (TPR)	97.67%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.80%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 4532	FN 108
Actual: Normal	FP 24	TN 12141

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	99.76%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	100.00%	Share of actual attacks successfully flagged
F1-Score	99.88%	Balanced measure combining Precision and Recall
Accuracy	99.76%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	45.83%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	0.00%	Actual attacks that went undetected
True Positive Rate (TPR)	100.00%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	54.17%	Normal traffic correctly identified as safe

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	0	0	0
Act: ICMP	0	0	0
Act: UDP	0	0	4532

2c. Hybrid Cascaded

Metric	Value	Description
Precision	99.47%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	97.67%	Share of actual attacks successfully flagged
F1-Score	98.56%	Balanced measure combining Precision and Recall
Accuracy	99.21%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.20%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	2.33%	Actual attacks that went undetected
True Positive Rate (TPR)	97.67%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.80%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 4532	FN 108
Actual: Normal	FP 24	TN 12141

2d. Response Latency

Metric	Value	Description
Detection Time	396.9 ms	Average time taken to detect an attack
Mitigation Response Time	1.6 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2e. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	1.13%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	7.01%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	6.83%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.21	5	4	2026-09-26 18:02:25	2026-09-26 18:05:03
10.0.0.20	4	3	2026-09-26 18:02:21	2026-09-26 18:05:02
10.0.0.22	4	3	2026-09-26 18:02:28	2026-09-26 18:05:06
10.0.0.8	2	2	2026-09-26 18:02:49	2026-09-26 18:03:40
10.0.0.4	1	1	2026-09-26 18:02:44	2026-09-26 18:02:54

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-26 18:02:21	10.0.0.20	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 18:02:25	10.0.0.21	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 18:02:28	10.0.0.22	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 18:02:32	10.0.0.20	DDoS	UDP Flood	79.4%	High	Time Ban (30s)
2026-09-26 18:02:32	10.0.0.20	DDoS	UDP Flood	79.4%	High	Time Ban
2026-09-26 18:02:35	10.0.0.21	DDoS	UDP Flood	79.2%	High	Time Ban (30s)
2026-09-26 18:02:35	10.0.0.21	DDoS	UDP Flood	79.2%	High	Time Ban
2026-09-26 18:02:38	10.0.0.22	DDoS	UDP Flood	86.3%	High	Time Ban (30s)
2026-09-26 18:02:39	10.0.0.22	DDoS	UDP Flood	86.3%	Critical	Time Ban
2026-09-26 18:02:49	10.0.0.8	DDoS	UDP Flood	92.4%	High	Time Ban (30s)
2026-09-26 18:02:54	10.0.0.4	DDoS	Uncertain	59.0%	Low	Released
2026-09-26 18:03:05	10.0.0.21	DDoS	UDP Flood	87.5%	High	Released
2026-09-26 18:03:15	10.0.0.21	DDoS	UDP Flood	87.5%	Critical	Time Ban (1m)
2026-09-26 18:03:19	10.0.0.8	DDoS	UDP Flood	92.4%	High	Released
2026-09-26 18:03:30	10.0.0.8	DDoS	UDP Flood	92.4%	High	Quarantined
2026-09-26 18:03:32	10.0.0.20	DDoS	UDP Flood	80.2%	High	Time Ban (1m)
2026-09-26 18:03:38	10.0.0.22	DDoS	UDP Flood	86.3%	Critical	Time Ban (1m)
2026-09-26 18:04:16	10.0.0.21	DDoS	UDP Flood	87.5%	Critical	Time Ban (2m)
2026-09-26 18:05:06	10.0.0.22	DDoS	UDP Flood	86.5%	Critical	Time Ban (2m)

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.22	UDP Flood	0.6261	86.5%	Critical	Phase 2	1	11s	2026-09-26 18:05:06	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6454	77.9%	Critical	Phase 2	1	11s	2026-09-26 18:05:03	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6329	79.1%	Critical	Phase 2	1	11s	2026-09-26 18:05:02	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6451	87.5%	Critical	Phase 2	4	1m 11s	2026-09-26 18:04:16	Attack persisted - escalating to time ban
10.0.0.8	UDP Flood	0.6224	92.4%	High	Phase 1	2	10s	2026-09-26 18:03:40	Attack Stopped
10.0.0.22	UDP Flood	0.6393	86.3%	Critical	Phase 2	3	1m 10s	2026-09-26 18:03:38	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.20	UDP Flood	0.6443	80.2%	High	Phase 2	3	1m 11s	2026-09-26 18:03:32	Attack persisted - escalating to time ban
10.0.0.8	UDP Flood	0.5757	92.4%	High	Phase 2	1	30s	2026-09-26 18:03:19	Ban Expired
10.0.0.21	UDP Flood	0.6355	87.5%	Critical	Phase 2	3	10s	2026-09-26 18:03:15	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6402	86.3%	High	Phase 2	2	40s	2026-09-26 18:03:08	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6059	87.5%	High	Phase 2	1	40s	2026-09-26 18:03:05	Ban Expired
10.0.0.20	UDP Flood	0.6463	80.2%	Critical	Phase 2	2	41s	2026-09-26 18:03:02	Attack persisted - escalating to time ban
10.0.0.4	Uncertain	0.6035	59.0%	Low	Phase 1	1	10s	2026-09-26 18:02:54	Attack Stopped
10.0.0.22	UDP Flood	0.6398	86.3%	High	Phase 2	1	10s	2026-09-26 18:02:38	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6471	79.2%	High	Phase 2	1	10s	2026-09-26 18:02:35	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6452	79.4%	High	Phase 2	1	11s	2026-09-26 18:02:32	Attack persisted - escalating to time ban

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE